

Devayush Rout

Applied AI Engineer

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Core Expertise

AI Reliability Engineering | AI Validation Infrastructure | Retrieval Systems | Agentic AI Systems | LLM Evaluation & Benchmarking | Document Intelligence

Professional Experience

AI Engineer Intern, Bynd AI

06/2026 – Present

Applied AI / Validation Infrastructure

Gurugram, Haryana

- Architected a production AI validation platform to evaluate LLM, RAG, document intelligence, and agentic systems for reliability, safety, and deployment readiness.
- Defined evaluation checks for hallucination detection, prompt-injection resilience, context grounding, instruction adherence, refusal behavior, ambiguity handling, and document fidelity.
- Built a validation engine that turns model outcomes into evidence-backed reports, readiness verdicts, explainable scores, and actionable remediation guidance for product teams.
- Developed automated benchmarking and regression infrastructure to catch quality regressions across model, prompt, retrieval, and workflow changes.

AI Builder Intern, FlowHouse (PrettiFlow)

05/2026 – 06/2026

AI Reliability Platform

Remote

- Built Truss, an AI reliability platform that evaluates LLM applications across correctness, robustness, safety, and production readiness.
- Designed evaluation pipelines for adversarial testing, hallucination detection, prompt-injection analysis, and comparative benchmarking across multiple language models.
- Researched evaluation methodology and contributed to product direction while selected for a 75+ builder AI infrastructure cohort.

AI Engineer Intern, FlyRank AI

01/2026 – 05/2026

Retrieval / RAG Infrastructure

Remote

- Architected retrieval infrastructure for enterprise AI applications using semantic search, metadata-aware filtering, deterministic routing, and vector retrieval.
- Built document ingestion, indexing, and RAG context-selection pipelines for enterprise document collections.
- Benchmarked embeddings, retrieval strategies, and search configurations to improve factual grounding in retrieval-augmented generation workflows.

Selected Engineering Projects

Internal Policy Intelligence System | *RAG, pgvector, evaluation, policy automation*

- Built an enterprise policy intelligence platform for HR, Finance, IT helpdesk, and compliance workflows using hybrid retrieval, metadata filtering, version-aware pgvector indexing, and deterministic routing.
- Implemented owner-scoped retrieval with SAFE/REFUSE/ESCALATE decisions and evaluated 100+ retrieval cases, reaching Recall@5 of 0.88 and 85% verdict accuracy.

Tender Match | *Hybrid retrieval, SQL filtering, tender intelligence*

- Designed an automated procurement intelligence platform that collects, filters, ranks, and matches government tenders to manufacturer capabilities.
- Combined SQL-based deterministic filtering with vector search to reduce manual tender review from about 2 hours/day to about 10 minutes.

MedWaste Guardian | *Computer vision, RAG, voice workflows*

- Built a multimodal AI assistant for biomedical waste compliance using computer vision, RAG-based regulation lookup, and voice guidance.
- Developed low latency interaction workflows for real-time disposal assistance and compliance reasoning in sterile clinical environments.

Technical Skills

Languages: Python, SQL

AI & LLM: LangChain, LlamaIndex, OpenAI API, Google Gemini, Ollama, PydanticAI, ClaudeCode SDK

Retrieval: pgvector, FAISS, ChromaDB, BM25

Backend: FastAPI, REST APIs | **Databases:** PostgreSQL, SQLite | **Infrastructure:** Docker, Git, Linux, Azure

Education

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

09/2023 – Present

Bhopal, MP, India